MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
Discipline Specific Elective (DSE) –MG5DSECHE301: GREEN CHEMISTRY FOR SUSTAINABLE DEVELOPMENT
MODEL QUESTION PAPER

Duration: 2 hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any ten questions
Each question carries 2 marks

1. What is the significance of Kyoto protocol? (K, CO-1)
2. How is the percentage yield of a reaction determined? (K, CO-1)
3. Name any two techniques used for waste minimization in chemical industries. (K, CO-2)
4. What is Atom Economy? (K, CO-1)
5. What is meant by Inherently Safer Design (ISD)? (K, CO-2)
6. What is MCPBA? List any one applications of MCPBA. (K, CO-3)
7. Compare the homogeneous and heterogeneous catalysis. (U, CO-2)
8. What is a supercritical fluid? Give one example used in Green Chemistry. (K, CO-3)
9. What are polymer-supported reagents? Mention their advantages. (K, CO-3)
10. Identify one environmental problem associated with the textile industry and suggest one green solution. (A, CO-4)
11. Explain the advantages of phase transfer catalysts. (U, CO-4)
12. Explain any one green method to synthesize adipic acid (U, CO-4)

[10 × 2 = 20]

Part B
Short Answer Questions
Answer any six questions
Each question carries 5 marks

13. Explain with examples why rearrangement reactions are considered to be 100 % atom economical. (U, CO-1)

14. Explain the Bhopal Gas tragedy and describe how inherently safer design (ISD) principles could have helped to reduce the associated hazards. (U, CO-2)
15. Illustrate the mechanism of photocatalysis with a neat and labeled diagram. (U, CO-2)
16. Explain the role of hydrogen peroxide as a green oxidant with suitable examples. (U, CO-3)
17. Explain the importance of the Rio, Montreal and Kyoto protocols in promoting environmental protection and sustainable development. (U, CO-1)
18. Write notes on Eco-friendly pesticides. (U, CO-4)
19. Apply the principles of green chemistry to select suitable green alternative to traditional solvents. (A, CO-3)
20. Apply the appropriate green wastewater treatment technologies to treat industrial wastewater and explain their advantages in terms of environmental sustainability. (A, CO-4).

[6 × 5 = 30]

Part C
Short Essay Questions
Answer any two questions
Each question carries 10 marks

21. Explain the principles of phase transfer catalysis to the synthesis of nitriles, alcohols, azides and alkyl fluorides from alkyl halides. Illustrate each transformation with suitable example (U, CO-4)
22. Explain the twelve principles of Green Chemistry. (U, CO-1)
23. Apply conventional heating, microwave heating and ultrasonic irradiation in chemical reactions. Compare their advantages and disadvantages in green synthesis. (A, CO-2)
24. Explain the use of oxygen, singlet oxygen, ozone, hydrogen peroxide and peroxy acids as green reagents. Compare their oxidation characteristics, and environmental advantages with suitable examples. (U, CO-3)

[2 × 10 = 20]
